R13

Code No: 118BA

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2023
DATABASE SECURITY
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

used paper July/Aug-2023

PART – A

(25 Marks)

1.a)	Define database inference with an example.	[2]
b)	What is the difference between Take rule and Grant Rule?	[3]
c)	Define the data confidentiality.	[2]
d)	Describe access Matrix Models.	[3]
e)	What is Modularity?	[2]
f)	Write about design pattern.	[3]
g)	What is the usage of frame-based system?	[2]
h)	Write the properties of object oriented data base.	[3]
i)	What does Jajodia model represent?	[2]
j)	List out some attributes of Orion model.	[3]

PART – B

(50 Marks)

2.a) Explain database vulnerability with a neat diagram.
b) Describe Take-Grant protection model in detail. [5+5]

OR

3.a) What is PN model? Write its features and advantages?
b) Write in detail about Acten model. [5+5]

4.a) Explain about Memory Protection, what is the need of MPU and MMU.
b) List out four gates of Operating System Security and explain in detail. [5+5]

OR

5.a) Explain control mechanism for resource protection.
b) Describe Lattice model for the flow control. [5+5]

6.a) Describe the statistical concepts in database protection.
b) Compare logical inference Vs statistical inference in the context of data base protection. [5+5]

OR

7.a) Explain different types of attacks on data base.
b) What is IDES system? Explain its functionality related to data base security. [5+5]

8.a) Discuss about multilevel database management system in detail.
b) Elaborate SORION protection model in detail. [5+5]

OR

9.a) Explain the security policy of MLS/ODBMS.
b) Describe the trusted computing base security policy in detail. [5+5]

10.a) Explain the ORION architecture in detail with a neat diagram.
b) What is the need of protection for active databases? Explain in detail. [5+5]

OR

11.a) Discuss about real-time security system in detail.
b) Explain the Jajodia model and its advantages. [5+5]

---ooOoo---

used paper July/Aug-2023